

Chapter 2

Topological Spaces

In this chapter we begin our study in earnest. The first order of business is to build up enough machinery to give a proper definition of manifolds. The chief problem with the provisional definition given in Chapter 1 is that it depends on having an “ambient Euclidean space” in which our n -manifold lives. This introduces a great deal of extraneous structure that is irrelevant to our purposes. Instead, we would like to view a manifold as a mathematical object in its own right, not as a subset of some larger space. The key concept that makes this possible is that of a *topological space*, which is the main topic of this chapter.

We begin by defining topological spaces, motivated by the open subset criterion for continuity in metric spaces. After the definition we introduce some of the important elementary notions associated with topological spaces such as closures, interiors, exteriors, convergence, continuity, and homeomorphisms, and then explore how to construct topologies from bases. At the end of the chapter we give the official definition of a manifold as a topological space with special properties.

Before you delve into this chapter, it would be a good idea to read quickly through the first two appendices to this book if you have not already done so. Much of the background material that is prerequisite for reading the first six chapters of this book is collected there.

Topologies

One of the most useful tools in analysis is the concept of a metric space. (See Appendix B for a brief review of metric space theory.) The most important examples, of course, are (subsets of) Euclidean spaces with the Euclidean metric, but many others, such as function spaces, arise frequently in analysis.

Our goal in this book is to study manifolds and those of their properties that are preserved by homeomorphisms (continuous maps with continuous inverses). To accomplish this, we could choose to view our manifolds as metric spaces. However, a metric still contains extraneous information. A homeomorphism between metric

spaces need not preserve distances (just think of the obvious homeomorphism between two spheres of different radii). So we will push the process of abstraction a step further, and come up with a kind of “space” without distances in which continuous functions still make sense.

The key idea behind the definition of this new kind of space is the open subset criterion for continuity (Theorem B.16, which you should review now). It shows that continuous functions between metric spaces can be detected knowing only the open subsets of both spaces. Motivated by this observation, we make the following definition. If X is a set, a **topology on X** is a collection $\mathcal{T}$ of subsets of X satisfying the following properties:

- (i) X and $\emptyset$ are elements of $\mathcal{T}$.
- (ii) $\mathcal{T}$ is closed under finite intersections: if $U_1, \dots, U_n$ are elements of $\mathcal{T}$, then their intersection $U_1 \cap \dots \cap U_n$ is an element of $\mathcal{T}$.
- (iii) $\mathcal{T}$ is closed under arbitrary unions: if $(U_\alpha)_{\alpha \in A}$ is any (finite or infinite) family of elements of $\mathcal{T}$, then their union $\bigcup_{\alpha \in A} U_\alpha$ is an element of $\mathcal{T}$.

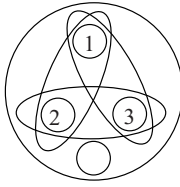
A pair $(X, \mathcal{T})$ consisting of a set X together with a topology $\mathcal{T}$ on X is called a **topological space**. Since we rarely have occasion to discuss any other type of space in this book, we sometimes follow the common practice of calling a topological space simply a **space**. As is typical in mathematics when discussing a set endowed with a particular kind of structure, if a particular choice of topology is understood from the context, we usually omit it from the notation and simply say “ X is a topological space” or “ X is a space.” Once X is endowed with a specific topology, the elements of X are usually called its **points**, and the sets that make up the topology are called the **open subsets of X** , or just **open sets** if both X and its topology are understood. With this terminology, the three defining properties of a topology can be rephrased as follows:

- X and $\emptyset$ are open subsets of X .
- Any intersection of finitely many open subsets of X is an open subset of X .
- Any union of arbitrarily many open subsets of X is an open subset of X .

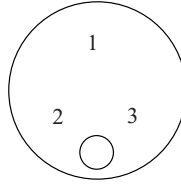
In this form, you will recognize these as the properties of open subsets of a metric space enumerated in Proposition B.5.

Aside from the simplicity of the open subset criterion for continuity, the other reason for choosing open subsets as the primary objects in the definition of a topological space is that they give us a qualitative way to detect “nearness” to a point without necessarily having a quantitative measure of nearness as we would in a metric space. If X is a topological space and $p \in X$, a **neighborhood of p** is just an open subset of X containing p . More generally, if $K \subseteq X$, a **neighborhood of the subset K** is an open subset containing K . (In some books, the word “neighborhood” is used in the more general sense of a subset containing an open subset containing p or K ; but for us neighborhoods are always open subsets.) We think of something being true “near p ” if it is true in some (or every, depending on the context) neighborhood of p .

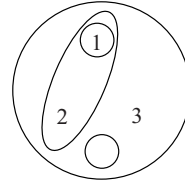
Here are some simple examples of topological spaces.



(a) Discrete topology



(b) Trivial topology



(c) $\{\{1\}, \{1, 2\}, \{1, 2, 3\}, \emptyset\}$

Fig. 2.1: Topologies on $\{1, 2, 3\}$.

Example 2.1 (Simple Topologies).

- (a) Let X be any set whatsoever, and let $\mathcal{T} = \mathcal{P}(X)$ (the **power set of X** , which is the set of all subsets of X), so every subset of X is open. (See Fig. 2.1(a).) This is called the **discrete topology on X** , and $(X, \mathcal{T})$ is called a **discrete space**.
- (b) Let Y be any set, and let $\mathcal{T} = \{Y, \emptyset\}$ (Fig. 2.1(b)). This is called the **trivial topology on Y** .
- (c) Let Z be the set $\{1, 2, 3\}$, and declare the open subsets to be $\{1\}$, $\{1, 2\}$, $\{1, 2, 3\}$, and the empty set (Fig. 2.1(c)). //

► **Exercise 2.2.** Verify that each of the preceding examples is in fact a topology.

Example 2.3 (The Metric Topology). Let (M, d) be any metric space, and let $\mathcal{T}$ be the collection of all subsets of M that are open in the metric space sense. It follows from Proposition B.5 that this is a topology, called the **metric topology on M** , or the **topology generated by d** . //

Metric spaces provide a rich source of examples of topological spaces. In fact, a large percentage of the topological spaces we need to consider are actually subsets of Euclidean spaces; since every such subset is a metric space in its own right (with the restriction of the Euclidean metric), it automatically inherits a metric topology. We call this the **Euclidean topology**, and unless we specify otherwise, subsets of $\mathbb{R}^n$ are always considered as topological spaces with this topology. Thus our intuition regarding topological spaces relies heavily on our understanding of subsets of Euclidean space.

Here are some standard subsets of Euclidean spaces that we will use throughout the book. Unless otherwise specified, each of these is considered as a topological space with the Euclidean (metric) topology.

- The **unit interval** is the subset $I \subseteq \mathbb{R}$ defined by

$$I = [0, 1] = \{x \in \mathbb{R} : 0 \leq x \leq 1\}.$$

- For any nonnegative integer n , the **(open) unit ball of dimension n** is the subset $\mathbb{B}^n \subseteq \mathbb{R}^n$ consisting of all vectors of length strictly less than 1:

$$\mathbb{B}^n = \{x \in \mathbb{R}^n : |x| < 1\}.$$

In the case $n = 2$, we sometimes call $\mathbb{B}^2$ the **(open) unit disk**.

- The **closed unit ball of dimension n** is the subset $\bar{\mathbb{B}}^n \subseteq \mathbb{R}^n$ consisting of vectors of length at most 1:

$$\bar{\mathbb{B}}^n = \{x \in \mathbb{R}^n : |x| \leq 1\}.$$

We sometimes call $\bar{\mathbb{B}}^2$ the **closed unit disk**.

- The **(unit) circle** is the subset $\mathbb{S}^1 \subseteq \mathbb{R}^2$ consisting of unit vectors in the plane:

$$\mathbb{S}^1 = \{x \in \mathbb{R}^2 : |x| = 1\}.$$

It is often useful to identify the plane $\mathbb{R}^2$ with the set $\mathbb{C}$ of complex numbers by the correspondence $(x, y) \leftrightarrow x + iy$, and think of the circle as the set of complex numbers with unit modulus:

$$\mathbb{S}^1 = \{z \in \mathbb{C} : |z| = 1\}.$$

We will use whichever representation is most convenient for the problem at hand.

- The **(unit) n -sphere** is the subset $\mathbb{S}^n \subseteq \mathbb{R}^{n+1}$ consisting of unit vectors in $\mathbb{R}^{n+1}$:

$$\mathbb{S}^n = \{x \in \mathbb{R}^{n+1} : |x| = 1\}.$$

Even though most of the topological spaces we consider are (subsets of) metric spaces with their metric topologies, it is important to recognize that not every topology can be defined by a metric. A topological space X is said to be **metrizable** if its topology happens to be the metric topology generated by some metric on X . Some spaces that are not metrizable are described in Example 2.36 and Problem 3-9.

If X is metrizable, the metric that generates its topology is not uniquely determined, because many different metrics can give rise to the same topology (meaning that the same sets are open with respect to both metrics). The next exercise describes a necessary and sufficient criterion for two metrics to generate the same topology, and some examples of such pairs of metrics.

► **Exercise 2.4.**

- Suppose M is a set and d, d' are two different metrics on M . Prove that d and d' generate the same topology on M if and only if the following condition is satisfied: for every $x \in M$ and every $r > 0$, there exist positive numbers r_1 and r_2 such that $B_{r_1}^{(d')}(x) \subseteq B_r^{(d)}(x)$ and $B_{r_2}^{(d)}(x) \subseteq B_r^{(d')}(x)$.
- Let (M, d) be a metric space, let c be a positive real number, and define a new metric d' on M by $d'(x, y) = c \cdot d(x, y)$. Prove that d and d' generate the same topology on M .
- Define a metric d' on $\mathbb{R}^n$ by $d'(x, y) = \max\{|x_1 - y_1|, \dots, |x_n - y_n|\}$. Show that the Euclidean metric and d' generate the same topology on $\mathbb{R}^n$. [Hint: see Exercise B.1.]
- Let X be any set, and let d be the discrete metric on X (see Example B.3(c)). Show that d generates the discrete topology.

- (e) Show that the discrete metric and the Euclidean metric generate the same topology on the set $\mathbb{Z}$ of integers.

Another important class of examples of topological spaces is obtained by taking open subsets of other spaces. If X is a topological space and Y is any open subset of X , then we can define a topology on Y just by declaring the open subsets of Y to be those open subsets of X that are contained in Y . The next exercise shows that this actually defines a topology on Y . (In the next chapter, we will see how to put a topology on *any* subset of a topological space.)

► **Exercise 2.5.** Suppose X is a topological space and Y is an open subset of X . Show that the collection of all open subsets of X that are contained in Y is a topology on Y .

► **Exercise 2.6.** Let X be a set, and suppose $\{\mathcal{T}_\alpha\}_{\alpha \in A}$ is a collection of topologies on X . Show that the intersection $\mathcal{T} = \bigcap_{\alpha \in A} \mathcal{T}_\alpha$ is a topology on X . (The open subsets in this topology are exactly those subsets of X that are open in each of the topologies $\mathcal{T}_\alpha$.)

Closed Subsets

Because of the importance of neighborhoods in understanding “nearness” and continuity in a topological space, the definition of a topological space takes open subsets as the primary objects. But there is a complementary notion that is almost as important.

If X is a topological space, a subset $F \subseteq X$ is said to be a **closed subset of X** if its complement $X \setminus F$ is an open subset. If X and its topology are understood, closed subsets of X are often just called **closed sets**. From the definition of topological spaces, several properties follow immediately:

- X and $\emptyset$ are closed subsets of X .
- Any union of finitely many closed subsets of X is a closed subset of X .
- Any intersection of arbitrarily many closed subsets of X is a closed subset of X .

A topology on a set X can be defined by describing the collection of closed subsets, as long as they satisfy these three properties; the open subsets are then just those sets whose complements are closed.

Here are some examples of closed subsets of familiar topological spaces.

Example 2.7 (Closed Subsets).

- (a) Any closed interval $[a, b] \subseteq \mathbb{R}$ is a closed subset of $\mathbb{R}$, as are the half-infinite closed intervals $[a, \infty)$ and $(-\infty, b]$.
- (b) Every closed ball in a metric space is a closed subset (Exercise B.8(a)).
- (c) Every subset of a discrete space is closed.
- (d) In the three-point space $\{1, 2, 3\}$ with the topology of Example 2.1(c), the closed subsets are $\emptyset$, $\{3\}$, $\{2, 3\}$, and $\{1, 2, 3\}$. //

It is important to be aware that just as in metric spaces, “closed” is not synonymous with “not open”—subsets can be both open and closed, or neither open nor closed. For example, in any topological space X , the sets X and $\emptyset$ are both open and closed subsets of X . On the other hand, the half-open interval $[0, 1)$ is neither open nor closed in $\mathbb{R}$.

Suppose X is a topological space and A is any subset of X . We define several related subsets as follows. The **closure of A in X** , denoted by $\bar{A}$, is the set

$$\bar{A} = \bigcap \{B \subseteq X : B \supseteq A \text{ and } B \text{ is closed in } X\}.$$

The **interior of A** , denoted by $\text{Int } A$, is

$$\text{Int } A = \bigcup \{C \subseteq X : C \subseteq A \text{ and } C \text{ is open in } X\}.$$

It follows immediately from the properties of open and closed subsets that $\bar{A}$ is closed and $\text{Int } A$ is open. To put it succinctly, $\bar{A}$ is “the smallest closed subset containing A ,” and $\text{Int } A$ is “the largest open subset contained in A .”

We also define the **exterior of A** , denoted by $\text{Ext } A$, as

$$\text{Ext } A = X \setminus \bar{A},$$

and the **boundary of A** , denoted by ∂A , as

$$\partial A = X \setminus (\text{Int } A \cup \text{Ext } A).$$

It follows from the definitions that for any subset $A \subseteq X$, the whole space X is equal to the disjoint union of $\text{Int } A$, $\text{Ext } A$, and ∂A . The set A always contains all of its interior points and none of its exterior points, and may contain all, some, or none of its boundary points.

For many purposes, it is useful to have alternative characterizations of open and closed subsets, and of the interior, exterior, closure, and boundary of a given subset. The following proposition gives such characterizations. Some of these are probably familiar to you from your study of Euclidean and metric spaces. See [Fig. 2.2](#) for illustrations of some of these characterizations.

Proposition 2.8. *Let X be a topological space and let $A \subseteq X$ be any subset.*

- (a) *A point is in $\text{Int } A$ if and only if it has a neighborhood contained in A .*
- (b) *A point is in $\text{Ext } A$ if and only if it has a neighborhood contained in $X \setminus A$.*
- (c) *A point is in ∂A if and only if every neighborhood of it contains both a point of A and a point of $X \setminus A$.*
- (d) *A point is in $\bar{A}$ if and only if every neighborhood of it contains a point of A .*
- (e) *$\bar{A} = A \cup \partial A = \text{Int } A \cup \partial A$.*
- (f) *$\text{Int } A$ and $\text{Ext } A$ are open in X , while $\bar{A}$ and ∂A are closed in X .*
- (g) *The following are equivalent:*

- *A is open in X .*

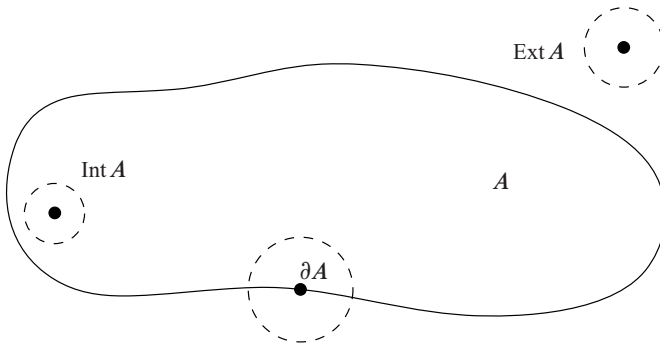


Fig. 2.2: Interior, exterior, and boundary points.

- $A = \text{Int } A$.
- A contains none of its boundary points.
- Every point of A has a neighborhood contained in A .

(h) The following are equivalent:

- A is closed in X .
- $A = \bar{A}$.
- A contains all of its boundary points.
- Every point of $X \setminus A$ has a neighborhood contained in $X \setminus A$.

► **Exercise 2.9.** Prove Proposition 2.8.

Given a topological space X and a set $A \subseteq X$, we say that a point $p \in X$ is a **limit point of A** if every neighborhood of p contains a point of A other than p (which itself might or might not be in A). Limit points are also sometimes called **accumulation points** or **cluster points**. A point $p \in A$ is called an **isolated point of A** if p has a neighborhood U in X such that $U \cap A = \{p\}$. Thus every point of A is either a limit point or an isolated point, but not both. For example, if $X = \mathbb{R}$ and $A = (0, 1)$, then every point in $[0, 1]$ is a limit point of A . If we let $B = \{1/n\}_{n=1}^{\infty} \subseteq \mathbb{R}$, then 0 is the only limit point of B , and every point of B is isolated.

► **Exercise 2.10.** Show that a subset of a topological space is closed if and only if it contains all of its limit points.

A subset A of a topological space X is said to be **dense in X** if $\bar{A} = X$.

► **Exercise 2.11.** Show that a subset $A \subseteq X$ is dense if and only if every nonempty open subset of X contains a point of A .

Convergence and Continuity

The primary reason topological spaces were invented was that they provide the most general setting for studying the notions of convergence and continuity. For this reason, it is appropriate to introduce these concepts next. We begin with convergence.

The definition of what it means for a sequence of points in a metric space to converge to a point p (see Appendix B) is really just a fancy way of saying that as we go far enough out in the sequence, the points of the sequence become “arbitrarily close” to p .

In topological spaces, we use neighborhoods to encode the notion of “arbitrarily close.” Thus, if X is a topological space, $(x_i)_{i=1}^{\infty}$ is a sequence of points in X , and $x \in X$, we say that **the sequence converges to x** , and **x is the limit of the sequence**, if for every neighborhood U of x there exists $N \in \mathbb{N}$ such that $x_i \in U$ for all $i \geq N$. Symbolically, this is denoted by either $x_i \rightarrow x$ or $\lim_{i \rightarrow \infty} x_i = x$.

► **Exercise 2.12.** Show that in a metric space, this topological definition of convergence is equivalent to the metric space definition.

► **Exercise 2.13.** Let X be a discrete topological space. Show that the only convergent sequences in X are the ones that are **eventually constant**, that is, sequences (x_i) such that $x_i = x$ for all but finitely many i .

► **Exercise 2.14.** Suppose X is a topological space, A is a subset of X , and (x_i) is a sequence of points in A that converges to a point $x \in X$. Show that $x \in \bar{A}$.

Next we address the most important topological concept of all: continuity. If X and Y are topological spaces, a map $f : X \rightarrow Y$ is said to be **continuous** if for every open subset $U \subseteq Y$, its preimage $f^{-1}(U)$ is open in X .

The open subset criterion for continuity in metric spaces (Theorem B.16) says precisely that a map between metric spaces is continuous in this sense if and only if it is continuous in the usual ε - δ sense. Therefore, all of the maps that you know from analysis to be continuous are also continuous as maps of topological spaces. Examples include polynomial functions from $\mathbb{R}$ to $\mathbb{R}$, linear maps from $\mathbb{R}^n$ to $\mathbb{R}^k$, and, more generally, every map from a subset of $\mathbb{R}^n$ to $\mathbb{R}^k$ whose component functions are continuous in the ordinary sense, such as polynomial, exponential, rational, logarithmic, absolute value, and trigonometric functions (where they are defined), and functions built up from these by composition.

Continuity can be detected by closed subsets as well as open ones.

Proposition 2.15. *A map between topological spaces is continuous if and only if the preimage of every closed subset is closed.*

► **Exercise 2.16.** Prove Proposition 2.15.

The next proposition gives some elementary but important properties of continuous maps. The ease with which properties like this can be proved is one of the virtues of defining continuity in terms of open subsets.

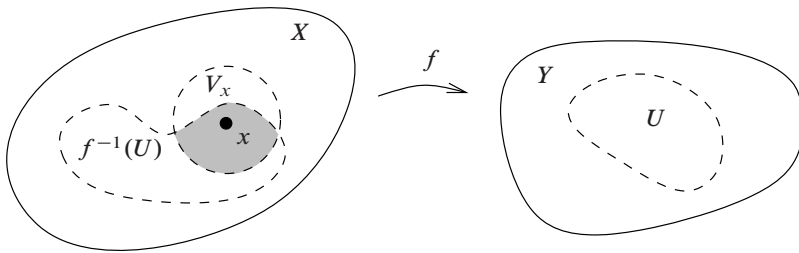


Fig. 2.3: Local criterion for continuity.

Proposition 2.17. *Let X , Y , and Z be topological spaces.*

- (a) *Every constant map $f : X \rightarrow Y$ is continuous.*
- (b) *The identity map $\text{Id}_X : X \rightarrow X$ is continuous.*
- (c) *If $f : X \rightarrow Y$ is continuous, so is the restriction of f to any open subset of X .*
- (d) *If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are both continuous, then so is their composition $g \circ f : X \rightarrow Z$.*

Proof. We prove (d) and leave the other parts as exercises. Suppose U is an open subset of Z ; we have to show that $(g \circ f)^{-1}(U)$ is an open subset of X . By elementary set-theoretic considerations, $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$. Applying the definition of continuity to g , we see that $g^{-1}(U)$ is an open subset of Y ; and then doing the same for f shows that $f^{-1}(g^{-1}(U))$ is an open subset of X . $\square$

► **Exercise 2.18.** Prove parts (a)–(c) of Proposition 2.17.

In metric spaces, one usually first defines what it means for a map to be *continuous at a point* (see Appendix B), and then a *continuous map* is one that is continuous at every point. In topological spaces, continuity at a point is not such a useful concept. However, it is an important fact that continuity is a “local” property, in the sense that a map is continuous if and only if it is continuous in a neighborhood of every point. The precise statement is given in the following important proposition.

Proposition 2.19 (Local Criterion for Continuity). *A map $f : X \rightarrow Y$ between topological spaces is continuous if and only if each point of X has a neighborhood on which (the restriction of) f is continuous.*

Proof. If f is continuous, we may simply take each neighborhood to be X itself. Conversely, suppose f is continuous in a neighborhood of each point, and let $U \subseteq Y$ be any open subset; we have to show that $f^{-1}(U)$ is open. Any point $x \in f^{-1}(U)$ has a neighborhood V_x on which f is continuous (Fig. 2.3). Continuity of $f|_{V_x}$ implies, in particular, that $(f|_{V_x})^{-1}(U)$ is an open subset of V_x , and is therefore also an open subset of X . Unwinding the definitions, we see that

$$(f|_{V_x})^{-1}(U) = \{x \in V_x : f(x) \in U\} = f^{-1}(U) \cap V_x,$$

so $(f|_{V_x})^{-1}(U)$ is a neighborhood of x contained in $f^{-1}(U)$. By Proposition 2.8(g), this implies that $f^{-1}(U)$ is an open subset of X . $\square$

If X and Y are topological spaces, a **homeomorphism from X to Y** is a bijective map $\varphi: X \rightarrow Y$ such that both φ and φ^{-1} are continuous. If there exists a homeomorphism between X and Y , we say that X and Y are **homeomorphic** or **topologically equivalent**. Sometimes this is abbreviated $X \approx Y$.

► **Exercise 2.20.** Show that “homeomorphic” is an equivalence relation on the class of all topological spaces.

The homeomorphism relation is the most fundamental relation in topology. In fact, as we mentioned in Chapter 1, *topological properties* are exactly those that are preserved by homeomorphisms. The next exercise shows, roughly speaking, that the topology is *precisely* the information preserved by homeomorphisms, and justifies the choice of topological spaces as the right setting for studying properties preserved by homeomorphisms. What this means in practice is that any property that can be defined purely in terms of open subsets will automatically be a topological property.

► **Exercise 2.21.** Let $(X_1, \mathcal{T}_1)$ and $(X_2, \mathcal{T}_2)$ be topological spaces and let $f: X_1 \rightarrow X_2$ be a bijective map. Show that f is a homeomorphism if and only if $f(\mathcal{T}_1) = \mathcal{T}_2$ in the sense that $U \in \mathcal{T}_1$ if and only if $f(U) \in \mathcal{T}_2$.

► **Exercise 2.22.** Suppose $f: X \rightarrow Y$ is a homeomorphism and $U \subseteq X$ is an open subset. Show that $f(U)$ is open in Y and the restriction $f|_U$ is a homeomorphism from U to $f(U)$.

It is also sometimes useful to compare different topologies on the same set. Given two topologies $\mathcal{T}_1$ and $\mathcal{T}_2$ on a set X , we say that **$\mathcal{T}_1$ is finer than $\mathcal{T}_2$** if $\mathcal{T}_1 \supseteq \mathcal{T}_2$, and **coarser than $\mathcal{T}_2$** if $\mathcal{T}_1 \subseteq \mathcal{T}_2$. The terminology is meant to suggest the picture of a subset that is open in a coarser topology being further subdivided into smaller open subsets in a finer topology.

► **Exercise 2.23.** Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be topologies on the same set X . Show that the identity map of X is continuous as a map from $(X, \mathcal{T}_1)$ to $(X, \mathcal{T}_2)$ if and only if $\mathcal{T}_1$ is finer than $\mathcal{T}_2$, and is a homeomorphism if and only if $\mathcal{T}_1 = \mathcal{T}_2$.

Here are a few explicit examples of homeomorphisms that you should keep in mind.

Example 2.24. Any open ball in $\mathbb{R}^n$ is homeomorphic to any other open ball; the homeomorphism can easily be constructed as a composition of **translations** $x \mapsto x + x_0$ and **dilations** $x \mapsto cx$. Similarly, all spheres in $\mathbb{R}^n$ are homeomorphic to each other. These examples illustrate that “size” is not a topological property. //

Example 2.25. Let $\mathbb{B}^n \subseteq \mathbb{R}^n$ be the unit ball, and define a map $F: \mathbb{B}^n \rightarrow \mathbb{R}^n$ by

$$F(x) = \frac{x}{1 - |x|}.$$

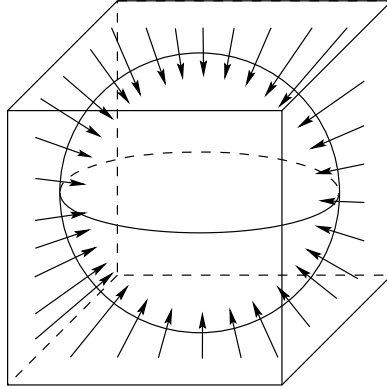


Fig. 2.4: Deforming a cube into a sphere.

Direct computation shows that the map $G: \mathbb{R}^n \rightarrow \mathbb{B}^n$ defined by

$$G(y) = \frac{y}{1 + |y|}$$

is an inverse for F . Thus F is bijective, and since F and $F^{-1} = G$ are both continuous, F is a homeomorphism. It follows that $\mathbb{R}^n$ is homeomorphic to $\mathbb{B}^n$, and thus “boundedness” is not a topological property. //

Example 2.26. Another illustrative example is the homeomorphism between the surface of a sphere and the surface of a cube alluded to in Chapter 1. Let $\mathbb{S}^2$ be the unit sphere in $\mathbb{R}^3$, and set $C = \{(x, y, z) : \max\{|x|, |y|, |z|\} = 1\}$, which is the cubical surface of side 2 centered at the origin. Let $\varphi: C \rightarrow \mathbb{S}^2$ be the map that projects each point of C radially inward to the sphere (Fig. 2.4). More precisely, given a point $p \in C$, $\varphi(p)$ is the unit vector in the direction of p . Thus φ is given by the formula

$$\varphi(x, y, z) = \frac{(x, y, z)}{\sqrt{x^2 + y^2 + z^2}},$$

which is continuous on C by the usual arguments of elementary analysis (notice that the denominator is always nonzero on C). The next exercise shows that φ is a homeomorphism. This example demonstrates that “corners” are not topological properties. //

► **Exercise 2.27.** Show that the map $\varphi: C \rightarrow \mathbb{S}^2$ is a homeomorphism by showing that its inverse can be written

$$\varphi^{-1}(x, y, z) = \frac{(x, y, z)}{\max\{|x|, |y|, |z|\}}.$$

In the definition of a homeomorphism, it is important to note that although the assumption that φ is bijective guarantees that the inverse map φ^{-1} exists for set-

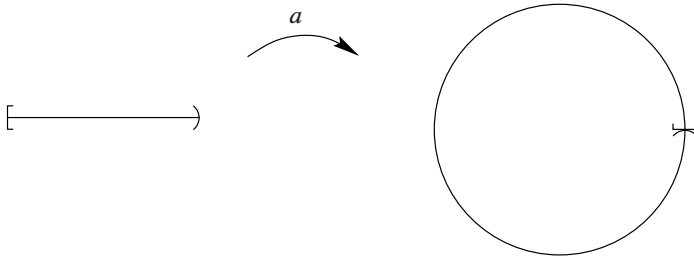


Fig. 2.5: A map that is continuous and bijective but not a homeomorphism.

theoretic reasons, continuity of φ^{-1} is not automatic. The next exercise gives an example of a continuous bijection whose inverse is not continuous.

► **Exercise 2.28.** Let X be the half-open interval $[0, 1) \subseteq \mathbb{R}$, and let $\mathbb{S}^1$ be the unit circle in $\mathbb{C}$ (both with their Euclidean metric topologies, as usual). Define a map $a: X \rightarrow \mathbb{S}^1$ by $a(s) = e^{2\pi i s} = \cos 2\pi s + i \sin 2\pi s$ (Fig. 2.5). Show that a is continuous and bijective but not a homeomorphism.

A map $f: X \rightarrow Y$ (continuous or not) is said to be an **open map** if it takes open subsets of X to open subsets of Y ; in other words, if for every open subset $U \subseteq X$, the image set $f(U)$ is open in Y . It is said to be a **closed map** if it takes closed subsets of X to closed subsets of Y . A map can have any of the properties “open,” “closed,” or “continuous” independently of whether it has the others (see Problem 2-5).

► **Exercise 2.29.** Suppose $f: X \rightarrow Y$ is a *bijective* continuous map. Show that the following are equivalent:

- (a) f is a homeomorphism.
- (b) f is open.
- (c) f is closed.

Proposition 2.30. Suppose X and Y are topological spaces, and $f: X \rightarrow Y$ is any map.

- (a) f is continuous if and only if $f(\bar{A}) \subseteq \overline{f(A)}$ for all $A \subseteq X$.
- (b) f is closed if and only if $f(\bar{A}) \supseteq \overline{f(A)}$ for all $A \subseteq X$.
- (c) f is continuous if and only if $f^{-1}(\text{Int } B) \subseteq \text{Int } f^{-1}(B)$ for all $B \subseteq Y$.
- (d) f is open if and only if $f^{-1}(\text{Int } B) \supseteq \text{Int } f^{-1}(B)$ for all $B \subseteq Y$.

Proof. Problem 2-6. □

There is a generalization of homeomorphisms that is often useful. We say that a map $f: X \rightarrow Y$ between topological spaces is a **local homeomorphism** if every point $x \in X$ has a neighborhood $U \subseteq X$ such that $f(U)$ is an open subset of Y and $f|_U: U \rightarrow f(U)$ is a homeomorphism.

Proposition 2.31 (Properties of Local Homeomorphisms).

- (a) *Every homeomorphism is a local homeomorphism.*
- (b) *Every local homeomorphism is continuous and open.*
- (c) *Every bijective local homeomorphism is a homeomorphism.*

► **Exercise 2.32.** Prove Proposition 2.31.

Hausdorff Spaces

The definition of topological spaces is wonderfully flexible, and can be used to describe a rich assortment of concepts of “space.” However, without further qualification, arbitrary topological spaces are far too general for most purposes, because they include some spaces whose behavior contradicts many of our basic spatial intuitions.

For example, in the spaces we are most familiar with, such as Euclidean spaces and metric spaces, a one-point set $\{p\}$ is always closed, because around every point other than p there is a ball that does not include p . More generally, two points in a metric space always have disjoint neighborhoods. However, these properties do not always hold in topological spaces. Consider the set $\{1, 2, 3\}$ with the topology of Example 2.1(c). In this space, 1 and 2 do not have disjoint neighborhoods, because every open subset that contains 2 also contains 1. Moreover, the set $\{1\}$ is not closed, because its complement is not open. And if that does not seem strange enough, consider the constant sequence $(2, 2, 2, \dots)$: it follows from the definition of convergence that this sequence converges both to 2 and to 3!

► **Exercise 2.33.** Let Y be a topological space with the trivial topology. Show that every sequence in Y converges to every point of Y .

The problem with these examples is that there are too few open subsets, so neighborhoods do not have the same intuitive meaning they have in metric spaces. In our study of manifolds, we want to rule out such “pathological” spaces, so we make the following definition. A topological space X is said to be a **Hausdorff space** if given any pair of distinct points $p_1, p_2 \in X$, there exist neighborhoods U_1 of p_1 and U_2 of p_2 with $U_1 \cap U_2 = \emptyset$. This property is often summarized by saying “points can be separated by open subsets.”

Example 2.34 (Hausdorff Spaces).

- Every metric space is Hausdorff: if p_1 and p_2 are distinct, let $r = d(p_1, p_2)$; then the open balls of radius $r/2$ around p_1 and p_2 are disjoint by the triangle inequality.
- Every discrete space is Hausdorff, because $\{p_1\}$ and $\{p_2\}$ are disjoint open subsets when $p_1 \neq p_2$.

- Every open subset of a Hausdorff space is Hausdorff: if $V \subseteq X$ is open in the Hausdorff space X , and p_1, p_2 are distinct points in V , then in X there are open subsets U_1, U_2 separating p_1 and p_2 , and the sets $U_1 \cap V$ and $U_2 \cap V$ are open in V , disjoint, and contain p_1 and p_2 , respectively. //

► **Exercise 2.35.** Suppose X is a topological space, and for every $p \in X$ there exists a continuous function $f: X \rightarrow \mathbb{R}$ such that $f^{-1}(0) = \{p\}$. Show that X is Hausdorff.

Example 2.36 (Non-Hausdorff Spaces). The trivial topology on any set containing more than one element is not Hausdorff, nor is the topology on $\{1, 2, 3\}$ described in Example 2.1(c). Because every metric space is Hausdorff, it follows that these spaces are not metrizable. //

These non-Hausdorff examples are obviously contrived, and have little relevance to our study of manifolds. But Problem 3-16 describes a space that would be a manifold except for the fact that it fails to be Hausdorff.

Hausdorff spaces have many of the properties that we expect of metric spaces, such as those expressed in the following proposition.

Proposition 2.37. *Let X be a Hausdorff space.*

- (a) *Every finite subset of X is closed.*
- (b) *If a sequence (p_i) in X converges to a limit $p \in X$, the limit is unique.*

Proof. For part (a), consider first a set $\{p_0\}$ containing only one point. Given $p \neq p_0$, the Hausdorff property says that there exist disjoint neighborhoods U of p and V of p_0 . In particular, U is a neighborhood of p contained in $X \setminus \{p_0\}$, so $\{p_0\}$ is closed by Proposition 2.8(h). It follows that finite subsets are closed, because they are finite unions of one-point sets.

To prove that limits are unique, suppose on the contrary that a sequence (p_i) has two distinct limits p and p' . By the Hausdorff property, there exist disjoint neighborhoods U of p and U' of p' . By definition of convergence, there exist $N, N' \in \mathbb{N}$ such that $i \geq N$ implies $p_i \in U$ and $i \geq N'$ implies $p_i \in U'$. But since U and U' are disjoint, this is a contradiction when $i \geq \max\{N, N'\}$. □

► **Exercise 2.38.** Show that the only Hausdorff topology on a finite set is the discrete topology.

Another important property of Hausdorff spaces is expressed in the following proposition.

Proposition 2.39. *Suppose X is a Hausdorff space and $A \subseteq X$. If $p \in X$ is a limit point of A , then every neighborhood of p contains infinitely many points of A .*

Proof. See Problem 2-7. □